

Broadband in-home networks

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The paper discusses the issue of broadband in-home networks. The purpose of the in-home network and the environment in which it has to operate are described. Architectural and technology options are evaluated and a possible solution is suggested. It is then shown how this solution can evolve as the bandwidth demands for applications increase. The status of the various residential broadband standards are also described with regard to the in-home network.

1. Introduction

With the advent of access network, terrestrial and satellite delivery [1] of digital entertainment and SOHO (small office home office) services [2], and with the increase in the number and use of home PCs, there will be a requirement for transporting a mixture of digital applications around the home. Customers will demand reliability and a service quality at least equal to, if not better than, that which they currently receive at the work place or from their domestic telephony/entertainment services. Therefore it is in BT's interests to ensure that these services are not only delivered to the network termination (NT) with a guaranteed quality of service (QoS), but also all the way to the customer premises equipment (CPE). Hence an in-home network (i.e. that part of the network which connects the NT to the CPE) must be designed which meets these new requirements from the point of view of both the service provider and the customer.

This in-home network must be capable of simultaneously carrying asynchronous data traffic and isochronous video/voice services. The user will want to initiate and control any of these services from anywhere on the in-home network. In order to provide this, the in-home network must be:

- symmetrical and full duplex,
- capable of carrying services to and between different rooms,
- able to deliver services without prior knowledge of location,
- independent of delivery mechanism,
- able to support different service types.

This implies that a 'general-purpose' in-home network is needed which is not tied to a single service or access technology.

Other key features of the in-home network are that it is highly cost-sensitive, and must not depend on skilled installation or configuration.

There are no obvious, existing, mature solutions which meet all of the above criteria. Therefore new architectures and topologies must be considered, and possibly specified within the relevant standards fora in order to enable mass production which will in turn drive down costs.

This paper discusses the issue of in-home distribution, focusing on the architectural and technology options, and suggests solutions which could evolve into a full functionality, general-purpose in-home network.

2. The purpose of the in-home network

The prime purpose of the in-home network is to provide distribution of services within the home, beyond the NT. There are two fundamentally different types of service — PC (personal computer) related data services (e.g. Internet access), and video entertainment. Therefore even an entry level home network should aim to support a minimum of two items of CPE simultaneously. It will then need to have a simple evolution path for the connection of more items/types of CPE, with the possible addition of new types of functionality such as intra-home communications.

From these considerations, the top-level user requirements for the home network can be defined as:

- support for a multiservice mix of video and data delivered from a variety of sources to be distributed to multiple locations in the home — note that this may produce an aggregate bandwidth of tens of megabits per second,
- intra-home communication to enable locally stored material (e.g. video recording) to be replayed, and to allow PC to PC/printer interconnection (i.e. to act as a home LAN).

In order for this to be a commercial success, it must be very simple to use from the customer perspective. Further system requirements are therefore:

- simple to install,
- simple to configure,
- robust,
- a single, universal interface standard is highly desirable.

It is likely that the standardisation of an open interface will be a requirement from the regulator, which will also act as an enabler for low-cost CPE.

2.1 In-home architectural assumption

One of the basic architectural assumptions is that the new broadband and existing narrowband home networks remain separate. Figure 1 shows a model of this general in-home architecture.

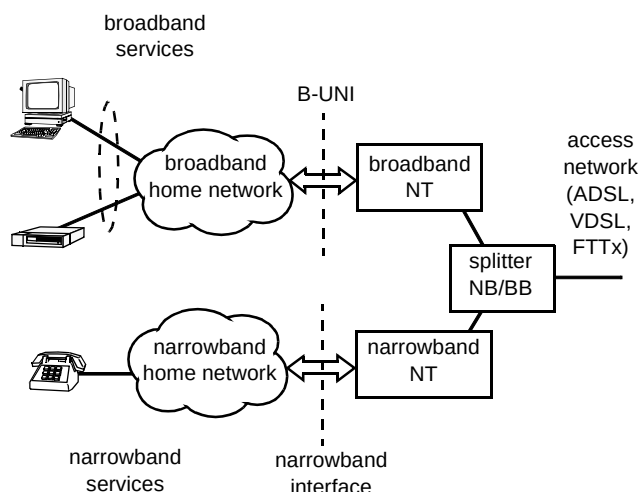


Fig 1 General architecture model based on xDSL and FTTx delivery.

A splitter, located within the home, separates the POTS from the broadband signals which are carried in separate cables within the home, with the narrowband network relying on existing cabling. It is not clear how the provision of additional narrowband services, e.g. a second phone line,

fits into this architecture, since the transport of narrowband services on the broadband network is not precluded, it simply must not depend solely upon it. This matter is currently being studied further.

2.2 Comparison with business architecture

Given this separation into narrowband and broadband networks, and the desire to have a general-purpose broadband home network, there might appear to be considerable similarity with the existing business LAN environment. Figure 2 shows a typical business configuration.

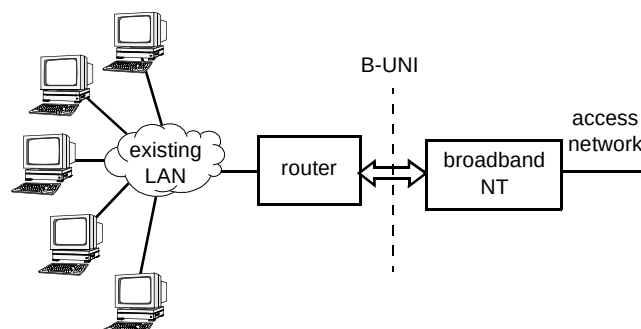


Fig 2 Typical business configuration.

The only obvious difference is the presence of a router which separates the corporate LAN (local area network) from the access network, providing firewall functionality and possibly a variety of interface types.

However, there are several significant differences between the home and business environments, which may mean that the same solutions are not directly applicable, namely:

- the bit rate and QoS demands are actually greater within the home, due to the greater importance of video-based entertainment services,
- the cost of the network equipment and CPE must be significantly less, even though it needs to support more demanding services,
- there is a legacy of cabling infrastructure in the business environment, which is not the case for the home,
- the aesthetics of both cabling and CPE are more important within the home,
- skilled installation and configuration cannot be assumed in the home,
- the electro-magnetic compatibility (EMC) requirements within the home are more stringent than in business premises.

Other significant differences are the need for greater security and management in the business environment.

3. The home environment

Two of the more significant differences will now be considered in greater detail.

3.1 EMC within the home

Electromagnetic compatibility concerns the ability of electronic equipment to function correctly in a disturbed environment (immunity), while limiting the disturbing effect (emission), of the equipment in its environment. EMC performance must be considered for an entire system (cabling, outlets and equipment).

For in-home networks, the more stringent Class B EMC regulations have to be met [3]. However, additional consideration must be given to noise ingress from sources peculiar to the domestic environment (e.g. washing machines and 13 amp plug insertions) which could cause burst errors, and thus significantly effect the performance of the in-home network transmission system. Similarly the prevention of interference from the in-home network (i.e. egress) with existing services is equally important. This is especially true if the interference affects a neighbour. Studies have shown that egress is more of a problem than ingress, with radio hams [4] and shortwave radio listeners being those most likely to experience interference.

However, with a suitable cabling and transmission system, the effects of both egress and ingress can be reduced to acceptable levels.

3.2 Installation and configuration issues

Due consideration of both installation and configuration issues is essential to ensure that a home network is socially and economically acceptable to customers.

To improve the aesthetics of an installation, any cabling used must blend into the surrounding decor, which has an impact on the cable colour and size. This also implies the need to fit cabling around existing room features, such as door frames, which can result in cable manufacturers' bend radius restrictions being exceeded, potentially jeopardising the transmission performance.

The size (and careful positioning) of the wall data jack units, the CPE and the NT equipment are also important and are non-trivial constraints as certain technologies are inherently bulky.

The PC is a special case of CPE as it will probably have been bought by the customer for another purpose, i.e. not just to connect to an access network. In most cases it will

need to be physically opened up and reconfigured in order to fit a network interface card to connect to the in-home network. The retrofitting of this card is a major issue for the telco, as it involves supporting numerous different types of PC with different hardware configurations and operating systems. Hence exploiting a pre-installed PC interface (e.g. Universal Serial Bus) may be a desirable alternative.

4. Infrastructure options

A decision needs to be made as to whether the in-home network is based on an existing, installed infrastructure type, or whether installing a new infrastructure is likely to be economically or socially acceptable. The existing installed wiring types in a typical home are considered first.

Telephone cabling is of fairly low quality and would need expensive transmission technology to support the bandwidths of interest; it also tends not to go to the places where the new services would be required. Power line (mains) distribution would again be difficult at high bandwidths, and there are security issues regarding not being able to constrain the signals within individual homes.

Co-axial cable for TV distribution is very common in certain parts of the world, notably the USA, but much less so in Europe and the Far East. There is some possibility of using this partially installed base, although in many cases newly installed co-axial cable would be required.

Coaxial cable does have very good transmission properties, significantly better than the twisted pair cable used in most LANs which is discussed below. It has lower attenuation, better controlled characteristic impedance, and does not suffer from crosstalk; duplex operation on coaxial cable is achieved by frequency division multiplexing (FDM).

EMC properties are particularly important in the domestic environment. Coaxial cable relies on a screen to shield the signals from the external environment. This works well at high frequencies, but is far less effective at low frequencies (a few MHz), which is unfortunate as this region is fairly noisy due to the effects of impulsive noise from domestic appliances, and radio amateur transmissions. There is also a maintenance problem as co-axial cable connectors are notorious for working loose.

However, the main concerns with co-axial cable are architectural and logistical. Cable TV distribution systems have a point to multipoint, tree-and-branch topology. This makes it very unsuitable as a full duplex system; splitters would probably need to be upgraded to support bi-directional operation, and a MAC protocol with an associated grant mechanism would be required to provide upstream communication in a controlled way. On the logistical side, co-axial cable is fairly difficult to install

because of the screen, and comes in unfortunate colours for home use, mainly black.

It therefore seems that there is no globally ubiquitous installed base which is suitable for broadband distribution. While there is no doubt that the cost and disruption of retrofitting a new infrastructure into existing homes is a valid concern, there are also considerable benefits in having a new, well-defined system. Putting in a well-controlled and possibly higher quality infrastructure, at some additional initial cost, means that all subsequent equipment attached to that infrastructure can be both simpler and cheaper.

The options for a new infrastructure can be broadly classified as fibre and copper.

The main attraction of fibre is its superior EMC properties. However, a system based on glass optical fibre would not be practical, given that unskilled installation must be possible, and installation rules such as bend radius would be ignored. The other alternative is plastic optical fibre (POF) which has the same advantages as glass, over a shorter link length, at a fraction of the cost. However, there are concerns with its installation robustness [5].

Turning to copper, the predominant cabling type used in the business LAN environment is unscreened twisted pair (UTP). This has gained acceptance for its logistical advantages — cost and ease of installation — in spite of it not being the best transmission medium.

UTP systems rely on the principle of balance rather than screening. The transmission systems used are differential; common mode interference is subtracted out at the receiver, and radiated interference cancels out. However, the degree to which this mechanism is effective depends on the cable balance; this is very good at low frequencies but decreases at higher frequencies. The screened and semi-screened variants of UTP (i.e. STP & FTP) also have the benefit of shielding, but have failed to become the general-purpose LAN cabling type due to cost, size and installation/connectorisation difficulties.

The only concern about the use of UTP in the home environment is whether unskilled installation would significantly affect the cable balance, and thereby give rise to transmission and EMC problems. Studies have shown that even very significant cable abuse, with regard to bending and cleating, has very little effect on the balance. Since the logistical advantages which have made UTP a business success are even more important in the home environment, UTP is a good choice for a newly installed home infrastructure. High-quality, category 5 UTP is little or no more expensive than its lower quality predecessor

(category 3), and gives significant EMC advantages, and so is the preferred type. It would be even more difficult to devise and implement a sound screening strategy in the home environment than in the business one, and so the screened TP variants are not appropriate.

There are two other options, a wireless solution, or some proprietary cabling scheme; the latter will be discussed further in section 5, as such cabling is linked to a particular transmission technology. At first sight a wireless solution seems very attractive as it avoids any new cabling, but it is doubtful whether current technologies meet the criteria of a general-purpose in-home network.

UTP combines adequate performance (> 100 Mbit/s) with relatively low cost, simple installation and mechanical robustness while supporting a variety of systems. These qualities make it ideal for the home network, but, to ensure maximum EMC effectiveness, category 5 UTP should be used.

5. Technologies

It is possible to take the view that the in-house distribution should be just a simple access network extension. The POTS signals are split off (in a so-called passive NT), and the access line transmission system is embedded in some item of CPE. Even for the case of a single attached item of CPE, there are interoperability and regulatory concerns with this approach, but the real difficulty comes when more than one attached device is required. This would require a point to-multipoint transmission system and MAC protocol, which would need to be replicated in each end device. The alternative approach is to terminate the access technology in an active NT, and then go into a separate home network. This has transmission, demarcation, planning and operational advantages, and can be shown to be cost effective for more than two items of CPE. The active NT approach is assumed in this paper, although interest in a passive NT has been rekindled by ADSL Lite, and the implications of this are also briefly considered.

There are a variety of technologies available today which could be used to connect the NT to several items of CPE. A detailed critique of each of these technologies is now given with the main contenders summarised in Table 1. The technologies can be categorised into three main areas, namely:

- existing business LANs,
- consumer electronic derived systems (USB and Firewire) requiring new, proprietary cabling,
- standardised or proprietary systems targeted at the home (CEBus) which aim to reuse existing wiring.

Table 1 Technology comparisons.

	Media and topology	Speed (Mbit/s)	Video QoS guaranteed
ATM 25/51	UTP FTP POF 50 m point-to-point	25.6 or 51.2	Yes
10BaseT	UTP 100 m shared star	10	No
100Base LANs	UTP 100 m shared star	100	Some
USB	STP 5 m tree and branch	12	Yes
Firewire	STP 4.5 m tree and branch	100, 200, 400	Yes

5.1 10 Mbit/s Ethernet

The UTP variant of Ethernet (i.e. 10BaseT) is commonly used within the business environment for local LAN connectivity. It is a shared access protocol which was designed for asynchronous data services, using up to 100 m of category 3 UTP. For a single point-to-point connection it is capable of transferring isochronous services at up to approximately 6 Mbit/s, but when used as a shared network there is no QoS support, and therefore it is unsuitable for video delivery.

The advantage of Ethernet is that it is widely available at a comparatively low cost in the business environment and therefore may be the most suitable interface for home working (e.g. using a laptop computer). However, due to the competitive market for general home PCs, the vast majority are not 'network ready' and do not have pre-installed 10BaseT network interface cards (NICs) which, as mentioned earlier, has implications for installation and configuration.

There is a major concern about the widespread use of 10BaseT in the home due to its poor EMC properties.

5.2 High-speed LANs

LANs with a bandwidth greater than 10 Mbit/s are becoming increasingly popular for business use. The technologies being adopted are typically 100 Mbit/s variants of Ethernet (100BaseTX, 100BaseVG), although there are others such as CDDI, 155 Mbit/s ATM and more recently Gigabit Ethernet. On cost and availability grounds, 100 Mbit/s Ethernet has the potential to be suitable for adoption within the home, but there is doubt whether the standard version is suitable for isochronous support, in spite of the increased speed. Although the EMC properties of these new systems are considerably better than 10BaseT, there is still a slight concern over the EMC performance.

100BaseVG does offer some isochronous support, but as a result it also has a more complex hub. It has also not been as commercially successful as other fast Ethernet technologies within the business environment.

The other systems mentioned above have more than enough bandwidth to cater for the services used within the home, but they are currently prohibitively expensive, and in some cases not suitable for isochronous support in a multi-user environment.

5.3 ATM

ATM (asynchronous transfer mode) is primarily designed to carry a mixture of both asynchronous and isochronous services simultaneously with a guaranteed QoS. ATM is a switched, point-to-point topology which can operate over TP or fibre. There are currently two ATM interfaces which are suitable for the in-home network, namely ATM 25 (25.6 Mbit/s) and ATM 51 (51.2 Mbit/s); the latter is a double-speed variant of ATM25 with which it is fully interoperable. This interface is currently being specified by the ATM Forum [6].

Since ATM is likely to be taken to the NT, there are several advantages in continuing ATM transport right into the various items of CPE:

- it avoids the cost and complexity of an interworking unit, or multiservice hub,
- it allows a single interface type, with cost and customer acceptance benefits,
- it allows flexible bandwidth delivery and hence flexible and upgradeable services,
- it permits a mixture of traffic types into a single item of CPE (e.g. a TV with Internet capabilities).

ATM was once viewed as the unifying LAN/WAN technology. It has not made significant in-roads into the business desktop, but the difference here is that the home environment actually needs the service mix which ATM can provide, whereas there is less of a case for this in the business world. Further, there is no installed and entrenched base of Ethernet in the residential market.

5.4 Wireless

Aesthetically wireless technology is the best choice for use within the home but terminal equipment is prohibitively expensive for high bandwidth systems, and it lacks both performance and functionality. It is available in two flavours, namely infra-red and radio.

2.4 GHz wireless LANs which are capable of supporting 2 Mbit/s are available now. There is also a developing

system called HiperLan (high performance radio LAN) which should be capable of supporting a payload of 12 Mbit/s. Products are not available today and will not be until approximately mid-1999. The system will have a data priority scheme that is intended to support isochronous services; however, QoS may not always be guaranteed as it depends on the vicinity of each installation.

Infra-red is another wireless technology which potentially has a greater bandwidth capability. Currently, there are a number of infra-red products available, supporting up to 10 Mbit/s. The dominant standard is now IrDA (infra-red data association) which supports 4 Mbit/s, but only over a very short range, line of sight, which would not even be sufficient for a single room. A cable backbone is also required to interconnect rooms. However, there are systems (not conforming to IrDA) which will support diffuse (i.e. non-line-of-sight) infra-red communications, but these are more limited in their capacity. Aligned systems offer greater bit rates, but are logistically impractical in the home.

5.5 Universal serial bus

The USB [7] was designed for traditional PC connections such as keyboards, mice, joysticks, etc, but also digital cameras, PC telephones and digital speakers are all currently available. The USB's data rate (12 Mbit/s) is more than adequate for many consumer applications including more advanced computer game devices, high-fidelity audio and highly compressed video, like MPEG-1 and MPEG-2. It has the ability to support a mix of asynchronous and isochronous services simultaneously, but there are concerns that, due to the limited bandwidth, the network could quickly become congested.

Another concern is that since USB was intended for the interconnection of a small cluster of equipment around the PC, it was designed to support a maximum cable length of only 5 m. This limit is fundamental to the system design and it is rather unlikely that a long reach USB will evolve.

However, in spite of this, USB does merit some consideration, possibly for an entry level system. The USB interface will add nothing to system cost as most PCs and its peripherals will come equipped with the USB as a standard interface (1998/99 timeframe) and it is currently supported by Windows®95. As mentioned earlier, eliminating the need for telcos to install and configure NICs is a major advantage.

5.6 IEEE1394 (Firewire)

Firewire [8] is similar to USB, but is targeted towards high-bandwidth consumer electronics connections to the PC. These include digital camcorders, cameras, digital

tools, satellite decoders and digital TV. Hence it operates with a much higher bit rate (100, 200 and 400 Mbit/s are available now, and 1.6 and 3.2 Gbit/s planned) and is more suited to carrying a large mix of services. Again Firewire was designed for the interconnection of clusters of audio and video equipment and consequently only has a cable reach of 4.5 metres. Line-powered repeaters are available, but expensive. Perhaps more beneficial to in-home networks is a proposal for a long reach Firewire using POF; however, this development is far from complete.

Firewire is around 1 to 2 years behind USB with respect to market acceptance and consequently it is not integrated into PCs as yet. It is also currently several times more expensive than USB.

5.7 CEBus (Home Plug 'n' Play)

CEBus was developed in the 1980s for the transfer of control and information sharing messages in order to automate the home, providing such functions as lighting control, temperature control, security control — all via a telephone. It is now known by an umbrella title called Home Plug 'n' Play. CEBus also has data channels as an optional extra which can carry audio, video and data files. It can also utilise the power line (mains) wiring to transfer control information only, or it can have its own dedicated infrastructure using coax or category 3 UTP cabling of up to 300 m between nodes. It is also possible to communicate using infra-red, and there are also plans for interoperability with Firewire.

CEBus is predominantly a low-bandwidth (up to 1 MHz) control system which uses a shared medium, and is unlikely to be suitable for the bit rate demands of future video services.

5.8 Impact of ADSL Lite

ADSL Lite is a lower cost version of ADSL (asynchronous digital subscriber line) which is currently generating a lot of interest within the industry. Basically it is a fairly high-speed ADSL modem with the POTS splitter removed, but it still supports the telephony service. It is intended to operate over the existing POTS infrastructure which exists within the home and will typically be in the form of a PC NIC. Standalone modems may also be available, connecting to the PC via USB.

The main concerns with this technology are that, as mentioned earlier, the existing in-home POTS infrastructure is of poor quality which may make the service susceptible to EMC problems. This cabling topology varies significantly from house to house, as this part of the network is unregulated. Also the POTS quality is degraded when the ADSL link is active and likewise the ADSL data rate is greatly reduced once a phone is in use. The positioning and

number of phones connected will also cause unpredictable results.

The drivers behind ADSL Lite are to reduce the cost of equipment and installation and while whole-life costs including support costs, and the reduced service capability need to be considered in any full business case, this thinking does have a knock-on effect on the home network. If ADSL Lite can be self-provisioned and reuse existing wiring, then these issues need to be re-addressed in the context of the home network. This has given rise to a resurgence of interest in distribution schemes which reuse existing phone wiring.

There are several proprietary systems becoming available in the market-place, predominantly in the USA. These are generally designed to either use the preinstalled POTS cabling or cable TV coax. Some systems use a stripped down version of some of the technologies mentioned above. They are intended to be very low cost devices in order to gain a quick market acceptance. However, at least some of these systems use proprietary carrier schemes which could seriously spectrally pollute the access network, and therefore compromise the delivery system. Such systems are also point-to-multipoint, shared media by definition, and so likely to be much more restricted with regard to the service mix and QoS which they can provide.

6. A solution for an in-home network

Assuming that the goal is still to provide a multiservice home distribution network, then the following approach has many advantages. The system should be based on end-to-end ATM over newly installed category 5 UTP. This configuration is very suitable for a telco which has ATM in the access network, as it ensures that ATM is delivered to the CPE; as a consequence QoS is controlled, and NT complexity is reduced due to the elimination of protocol conversion.

However, there are two concerns with this approach — how to attach to a significant number of end devices and how to accommodate other network types. Although a single ubiquitous network type is the goal, there is one other network type which cannot be completely ignored.

The only current way to provide multiple attachments would be by means of a full ATM switch which is prohibitively expensive; while there are development examples of very low cost ATM hubs, these are not yet commercially available. The main problem is with providing the second connection, as this meets the immediate needs of the two service types (data and entertainment), and establishes the concept of simultaneous multiservice delivery with both the customer and equipment

provider, which may accelerate the development of low-cost hubs.

One possible way of achieving this is shown in Fig 3. There are two ATM interfaces on the NT which allows two simultaneous connections from the outset. Note, however, that there must be sufficient functionality in (and control of) the NT to allow QoS management. This approach is analogous to current ISDN2 delivery, where two connections are presented. If there is a desire to attach more than two ATM devices then an additional small hub would be required.

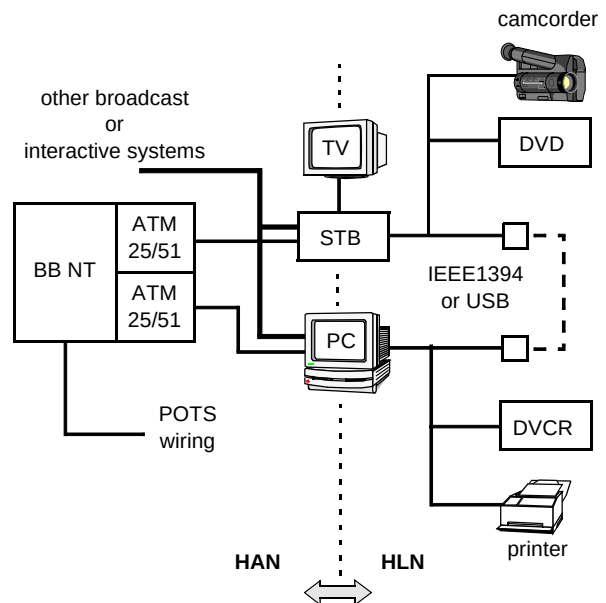


Fig 3 Typical in-home network.

With regard to other network types, the main consideration is co-existence with IEEE1394, which is starting to appear in some high-end consumer devices. The home network shown is split into two distinct sections, the home access network (HAN) whose main purpose is the connection of access network delivered services to various items of CPE, and the home local network (HLN).

The HLN is based on IEEE1394 (or possibly USB) and has evolved from the consumer electronic manufacturers' requirements for the interconnection of local clusters of audio/video equipment.

The interface between the HAN and HLN will be via customer-owned equipment (e.g. a PC) which is capable of supporting both networks.

Depending on the implementation, the telco NT may also support intra-home communications, i.e. device-to-device communication within the home without going via the access network, but local regulatory restrictions might

prevent such functionality being incorporated into the NT itself. Therefore intra-home support could either be solely the function of the HLN, which is inherent to the technology used, or a small low-cost ATM hub could be placed between the NT and the CPE, thus providing an extra layer of switching.

7. Standardisation

The in-home network is currently being developed in several standards and discussion forums. Below is a brief description of the activities of the major players, in which BT has been an active participant. BT has also attempted to ensure that there is cohesion between the various bodies wherever possible.

7.1 ATM Forum

The Residential Broadband (RBB) Working Group is currently developing two specifications [6, 9] which define a complete end-to-end solution for ATM delivery to the CPE, and also the physical specification for the ATM25/51 in-home network interface.

BT has driven and influenced much of this work, and currently chairs the working group.

7.2 DVB (Digital Video Broadcasting)

The DVB currently have a specification [10] for an in-home digital network based on IEEE1394 and an ATM NT, similar to that shown in Fig 3 above. It is proposed that this specification is made into an ETSI standard. Again BT has driven and influenced much of this specification.

7.3 FSAN (Full Service Access Network)

FSAN [11] is a consortium of predominantly European telcos of which BT is a member. They are in the process of developing a customer network specification [12] which can be connected to an xDSL or FTTx access network.

7.4 DAVIC (Digital Audio Visual Council)

DAVIC [13] is covering similar ground to the ATM Forum, although this is just a part of a much larger consideration of complete end-to-end delivery. DAVIC has come up with a set of scenarios which are very similar to those of the ATM Forum.

8. Conclusions

In order to provide a true, multiservice distribution capability, the in-home network needs to be a general-purpose distribution system, using an active NT, based on point-to-point links transporting ATM cells. The preferred medium is category 5 UTP cable, although the use of POF might be sensible in troublesome EMC environments. The NT could have two home network interfaces allowing connection to two items of CPE from the outset. These interfaces will be capable of carrying at least 25 Mbit/s but ideally 50 Mbit/s. When the home network needs to support more than two items of CPE, it can either evolve to a hub-based architecture, or it can utilise non-ATM-based technologies such as Firewire or USB. The above are all currently being developed in various standards bodies and discussion groups, in which BT is very active

Acknowledgements

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In 1989 he graduated with a 1st class honours (BEng) degree and subsequently moved to BT Laboratories.

Initially he conducted the world's first extensive survey of impulsive noise in the access network. More recently he has moved into the private data network area which has involved the practical performance testing of business LANs, voice and data integration studies, and the development of restricted broadband networks.



David Thorne holds a BSc in Physics from Imperial College, London, and a PhD in Radio Astronomy from the University of Manchester. He began his career with Marconi Elliott Avionics, and then moved briefly to ICL before embarking on 10 years in the instrument science division of the Royal Greenwich Observatory. He joined BT in 1989, initially working on the development of novel fault-finding techniques for the access network. He then moved into the private data network area and currently leads a team which is concerned with all aspects of (business) local area networks, ranging from

their performance to the specification and requirements of the supporting infrastructure. More recently he has become very involved in studies concerning the distribution of network-delivered broadband services within the residential and small business environments. He chairs the ATM Forum Residential Broadband (RBB) Working Group, which is defining specifications for the delivery and distribution of ATM-based residential services.